

SMS Notification System for Potential Radiological Risk Scenarios

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1. INTRODUCTION

Safety in Nuclear Engineering relies on the continuous monitoring of critical systems to prevent radiological and nuclear risks. In this context, multiple detection and monitoring mechanisms are implemented to mitigate the risks associated with ionizing radiation [3]. This work proposes the development of a Control System, implemented on a Raspberry Pi platform, capable of integrating ionizing radiation detectors with an SMS messaging API. The system's architecture allows a predefined dose rate threshold (setpoint) to be configured, such that, when exceeded, an automatic notification is sent to a group of previously registered users.

2. MATERIALS AND METHODS

For this experiment, the embedded system consisted of a Raspberry Pi 3B controller and a BG51 semiconductor radiation detector, as illustrated in Figure 1.

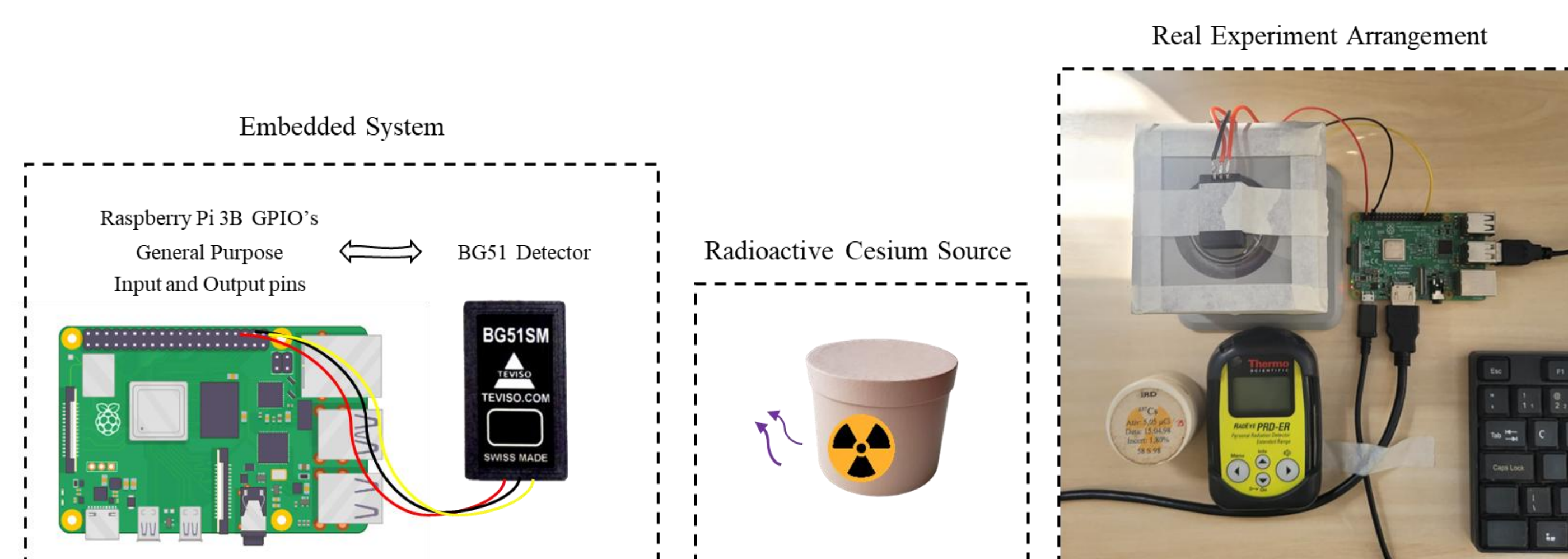


Figure 1 - Diagram of the Experiment

Source: The Authors (Own).

The choice of the Raspberry Pi is justified by its convenience as a controller: it has power and communication inputs that allow for the straightforward integration of the BG51 detector via GPIO (General Purpose Input and Output) pins, in addition to a built-in WiFi module, which enables the implementation of SMS messaging through a web application using Twilio.

As for the selected detector, its choice is supported by its low power consumption (25μA), making it ideal for applications in radiation-harsh environments, requiring little to no maintenance over extended periods of operation. Furthermore, the detector can detect gamma and beta radiation, perform energy discrimination, and offers a wide operational range between 70 keV and 2 MeV, with a dose rate measurement range spanning from 0.1 μSv/h to 100 mSv/h.

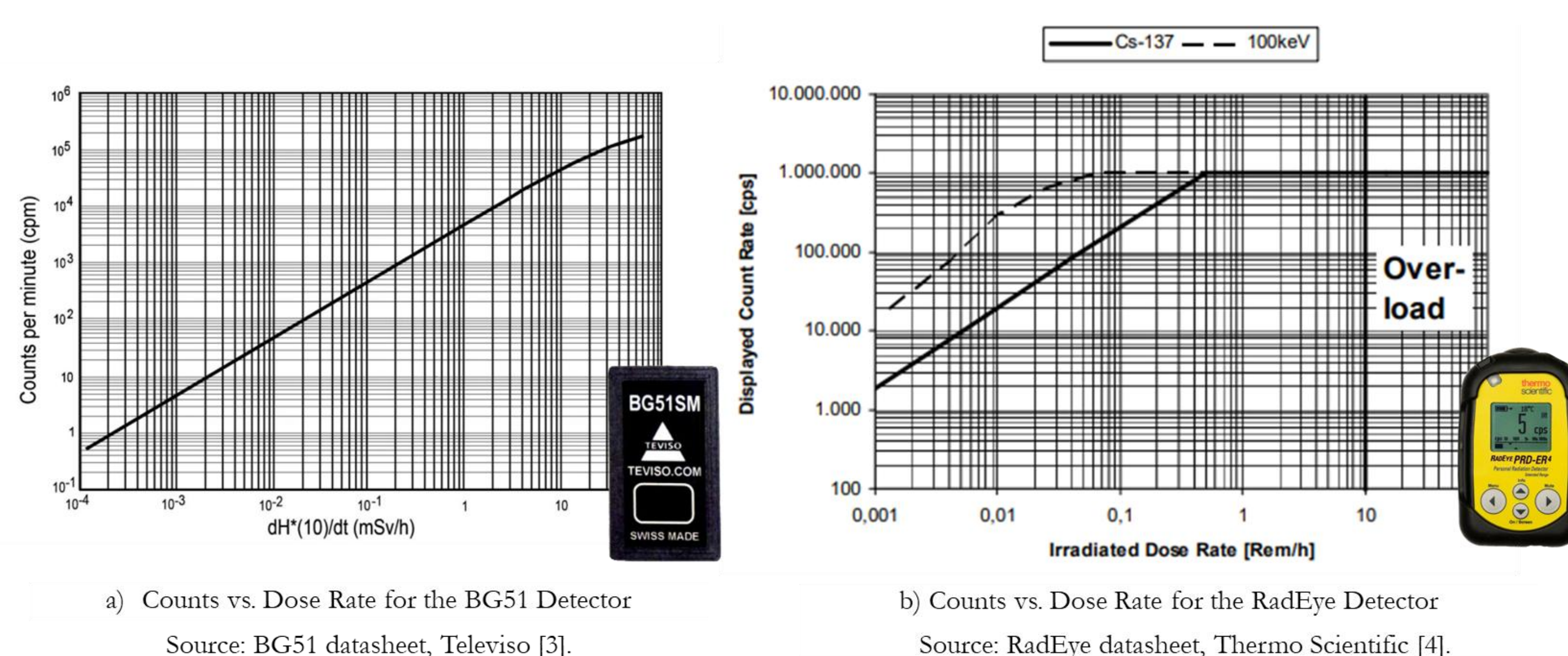


Figure 2 – Counts vs. Dose Rate Graphs

Based on the graph in Figure 2 and the information from the BG51 datasheet, the detector's conversion factor was identified, establishing a relationship of 5 CPM for each 1 μSv/h of dose rate. To apply this factor, the dose calculation methodology adopted was based on the number of events per minute (CPM) recorded by the system [1].

3. RESULTS

When the radioactive sample was placed near the detector, an increase in the dose rate was observed, as expected. This increase activated the function that executes the SMS sending script, which successfully sent a message, via Twilio API, to a previously selected phone number containing information about the radiation measured and processed by the embedded system, as illustrated in Figure 3.

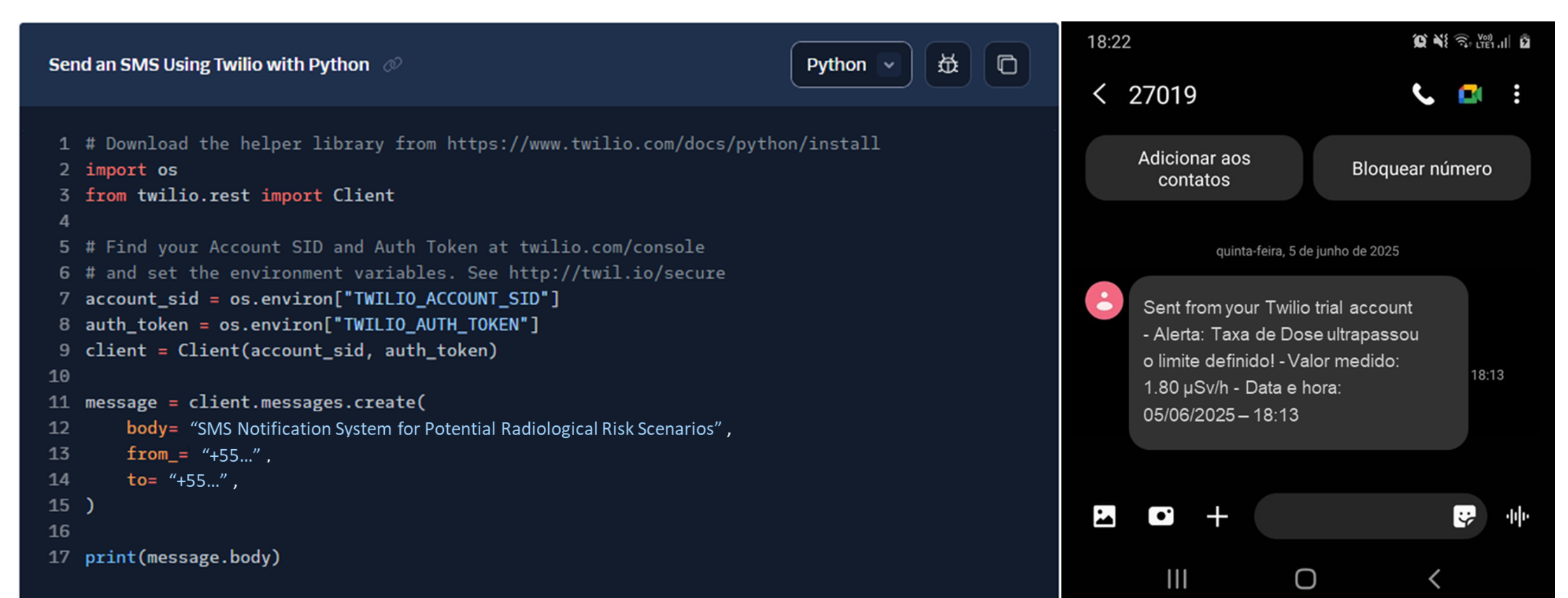


Figure 3 – (a) Python Script for Sending SMS and (b) Experiment SMS screenshot

Source: Twilio Documentation; The Authors (Own).

4. CONCLUSIONS

The proposed SMS-based detection and notification system proved effective in alerting a designated user when radiation levels exceeded the trigger threshold. The application also allows customization of the message body, enabling the inclusion of details such as dose rate, date, time, and location to inform professionals, response teams, or the public. However, limitations related to service quality and system availability must be addressed to ensure reliable and consistent message delivery.

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